INTRODUCTION TO NATURAL LANGUAGE PROCESSING

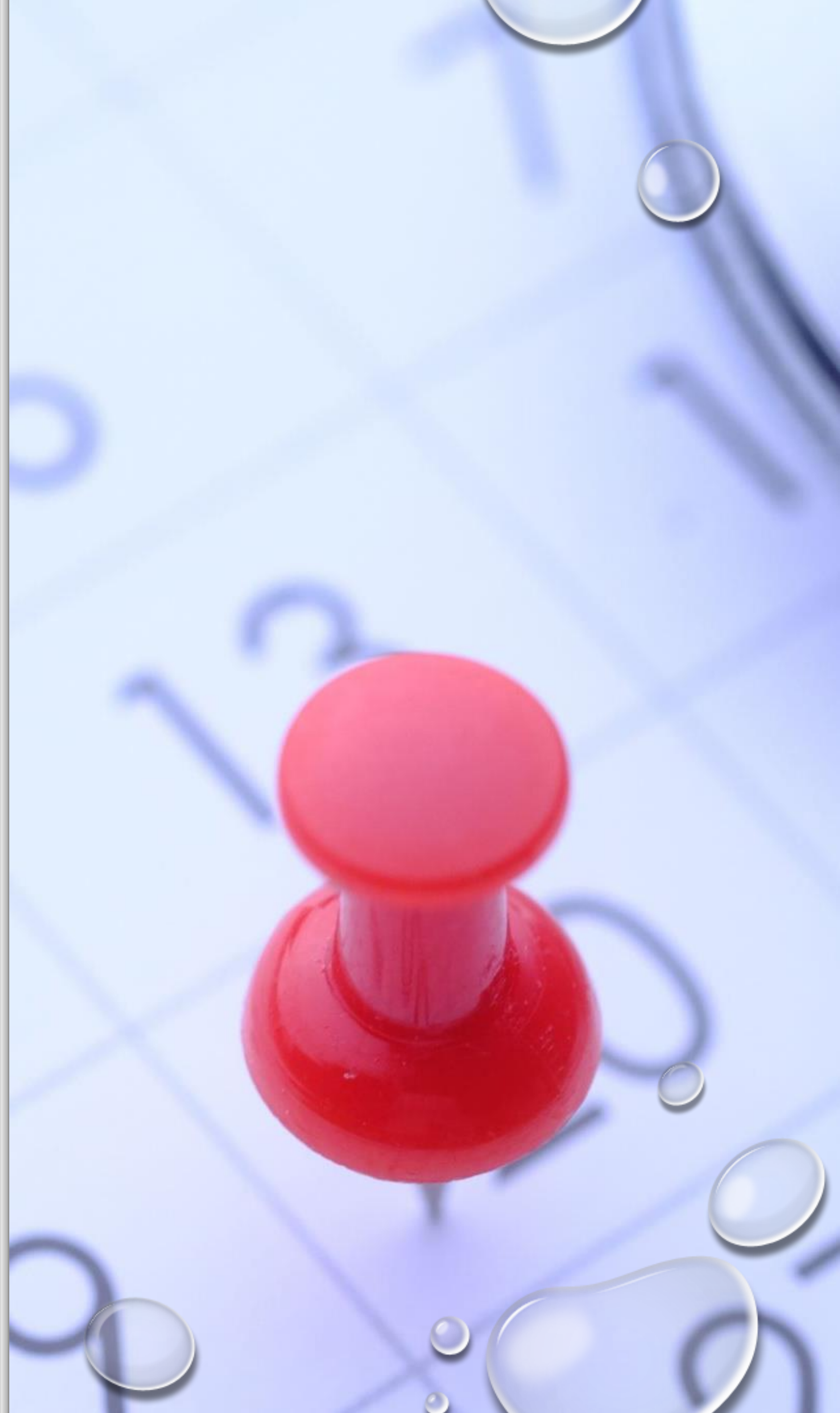
MANISH SHRIVASTAVA

OUTLINE

- Logistics
- Overview
- Projects
- Discussion

LOGISTICS

- FAST-PACED, HANDS-ON
- FROM BASICS TO STATE-OF-THE-ART
- QUIZ, ASSIGNMENTS, PROJECTS
- TIMING 8:30 – 9:55 WEDNESDAY, SATURDAY



WEIGHTAGES

- Assignments: 20% (5 + 5 + 10)
- Quizzes: 15% (7.5 + 7.5)
- Project: 45%
- Final Exam: 20%

ASSIGNMENTS (20%)

#	Release	Due
1	21 January	11 February
2	12 February	5 March
3	6 March	27 March

Strict deadlines. No exceptions for deadline extension
Grade will be normalized before release

QUIZZES (15%) & EXAM (20%)

- DETAILS WILL BE ANNOUNCED

PROJECT (45%)

- PROPOSE CUSTOM PROJECTS
- TEAM SIZE: 3 OR 4
- RANDOM TEAM ALLOTMENT OPTION AVAILABLE
- RESUBMIT PROPOSAL IF REJECTED
- UTILIZE TA OFFICE HOURS TO DISCUSS PROJECTS FEASIBILITY* AND SCOPE

* Compute cannot be provided :(

PROJECT EXPECTATION

- INTERMEDIATE/ADVANCED LEVEL PROJECTS
  - HIGHLY ENCOURAGED TO SUBMIT WORK TO CONFERENCES/WORKSHOPS
- PROJECTS NOT SUITABLE:
  - FINETUNING/INSTRUCTION FINETUNING
  - RAG

PROJECT EXPECTATION

- SUITABLE PROJECTS
  - ARCHITECTURE OPTIMIZATION & CHANGES
  - NOVEL EVALUATION METHODS
  - DATASETS*
  - NOVEL TASKS

PROJECT AREAS

- Efficient/Low-Resource Methods
- Ethics, Fairness, Bias
- Generation/Language Modeling
- Interpretability/Explainability
- Multilingual/Cross-lingual
- Multimodal
- NLP Applications
- Evaluation/Benchmarking
- Retrieval & Extraction
- Document Understanding
- Graph + LLMs
- Entertainment + LLM

PROJECT TIMELINE

Item	Deadline
Team Details/Assignment Form	**17 January**
Interim Project Proposal	**20 January**
Proposal Rejection by TAs	22 January
Project Finalization, Mentor Assignment	26 January
Final Proposal	**3 February**
Mid Submission	**8 March**
Final Submission	**8 April**

COURSE WEBSITE

https://ltrcnlp.github.io/S26/

COURSE OBJECTIVES

- Demonstrate the knowledge of stages and fundamental building blocks of NLP
- Apply NLP machine learning algorithms for classification, representation, and parsing
- Demonstrate the knowledge of Dense vector representation for NLP
- Explain the concepts behind distributed semantics
- Discuss the approaches to global and contextual semantic representation
- Apply the above concepts for fundamental NLP tasks.

WHY NLP?

- UBIQUITOUS!!
- MULTI-FACETED
- NEED OF THE HOUR
- UNSOLVED ☺

SYLLABUS

Unit 1: Stages of NLP: from lexical to semantic. Fundamental Language processing : Tokenization, Language modeling, Text classification,

Unit 2: Morphology, POS Tagging, Chunking, Discriminative vs generative modes, HMM and CRF

Unit 3: Syntax parsing: Constituency and Dependency, PCFG, projectivity Arc-eager

Unit 4: Distributed semantics: SVD, Word2Vec, RNN, LSTM,

Unit 5: Contextual Distributed semantics: ElMO, BERT

INTRODUCTION TO NLP

MANISH SHRIVASTAVA

WHAT IS NLP?

- ALSO CALLED COMPUTATIONAL LINGUISTICS
  - NOT REALLY THE SAME
- PROCESSING OF LANGUAGE CONTENT
  - TEXT AND SPEECH
  - IDENTIFYING LANGUAGE PHENOMENA (GRAMMAR ETC.)
  - UNDERSTANDING LANGUAGE CONTENT (MEANING, INTENT, SENTIMENT ETC.)

WHAT IS NLP

- **WIKI: NATURAL LANGUAGE PROCESSING** (**NLP**) IS A FIELD OF COMPUTER SCIENCE, ARTIFICIAL INTELLIGENCE, AND COMPUTATIONAL LINGUISTICS CONCERNED WITH THE INTERACTIONS BETWEEN COMPUTERS AND HUMAN (**NATURAL**) LANGUAGES.

GO BEYOND THE KEYWORD MATCHING

- IDENTIFY THE STRUCTURE AND MEANING OF WORDS, SENTENCES, TEXTS AND CONVERSATIONS
- DEEP UNDERSTANDING OF BROAD LANGUAGE
- NLP IS ALL AROUND US

TWO GENERATIONS OF NLP

- HAND-CRAFTED SYSTEMS – KNOWLEDGE ENGINEERING [1950S–]
  - RULES WRITTEN BY HAND; ADJUSTED BY ERROR ANALYSIS
  - REQUIRE EXPERTS WHO UNDERSTAND BOTH THE SYSTEMS AND DOMAIN
  - ITERATIVE GUESS-TEST-TWEAK-REPEAT CYCLE

- AUTOMATIC, TRAINABLE (MACHINE LEARNING) SYSTEM [1985S–]
  - THE TASKS ARE MODELED IN A STATISTICAL WAY
  - MORE ROBUST TECHNIQUES BASED ON RICH ANNOTATIONS
  - PERFORM BETTER THAN RULES (PARSING 90% VS. 75% ACCURACY)

- UNSUPERVISED SEMANTICS [2011-]
  - DEEP LEARNING

NLP

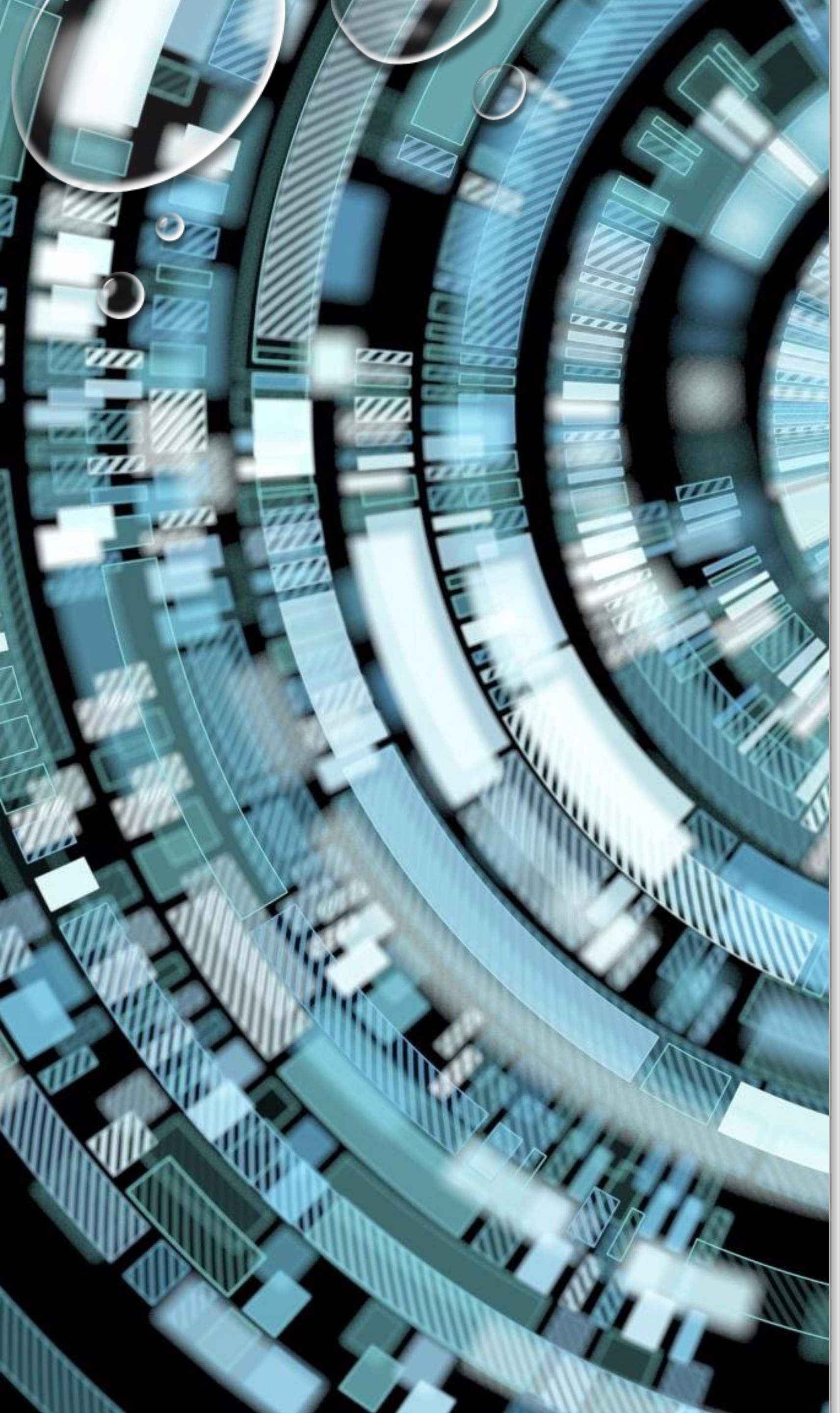
Computers use (analyze, understand, generate) natural language

Text Processing

- Lexical: tokenization, part of speech, head, lemmas
- Parsing and chunking
- Semantic tagging: semantic role, word sense
- Certain expressions: named entities
- Discourse: coreference, discourse segments

Speech Processing

- Phonetic transcription
- Segmentation (punctuations)
- Prosody



TRENDS

- AN ENORMOUS AMOUNT OF KNOWLEDGE IS NOW AVAILABLE IN MACHINE READABLE FORM AS NATURAL LANGUAGE TEXT
- CONVERSATIONAL AGENTS ARE BECOMING AN IMPORTANT FORM OF HUMAN-COMPUTER COMMUNICATION
- MUCH OF HUMAN-HUMAN COMMUNICATION IS NOW MEDIATED BY COMPUTERS

MAJOR LEVELS

1. WORDS
2. SYNTAX
3. MEANING
4. DISCOURSE

5. Applications exploiting each

CHALLENGES – AMBIGUITY

- WORD SENSE / MEANING AMBIGUITY

Credit: http://stuffsirisaid.com

CHALLENGES – AMBIGUITY

- PP ATTACHMENT AMBIGUITY

San Jose cops kill man with knife

Text Paper Translate Listen Close

San Jose cops kill man with knife

Ex-college football player, 23, shot 9 times allegedly charged police at fiancee's home

By Hamed Aleaziz and Vivian Ho

A man fatally shot by San Jose police officers while allegedly charging at them with a knife was a 23-year-old former football player at De Anza College in Cupertino who was distraught and depressed, his family said Thursday.

Police officials said two officers opened fire Wednesday afternoon on Phillip Watkins outside his fiancee's home because they feared for their lives. The officers had been drawn to the home, officials said, by a 911 call reporting an armed home invasion that, it turned out, had been made by Watkins himself.

But the mother of Watkins' fiancee, who also lives in the home on the 1300 block of Sherman Street, said she witnessed the shooting and described it as excessive. Faye Buchanan said the confrontation happened shortly after she called a suicide intervention hotline in hopes of getting Watkins medical help.

Watkins' 911 call came in at 5:01 p.m., said Sgt. Heather Randol, a San Jose police spokeswoman. "The caller stated there was a male breaking into his home armed with a knife," Randol said. "The caller also stated he was locked in an upstairs bedroom with his children and requested help from police."

She said Watkins was on the sidewalk in front of the home when two officers got there. He was holding a knife with a 4-inch blade and ran toward the officers in a threatening manner, Randol said.

"Both officers ordered the suspect to stop and drop the knife," Randol said. "The suspect continued to charge the officers with the knife in his hand. Both officers, fearing for their safety and defense of their life, fired at the suspect."

On the police radio, one officer said, "We have a male with a knife. He's walking toward us."

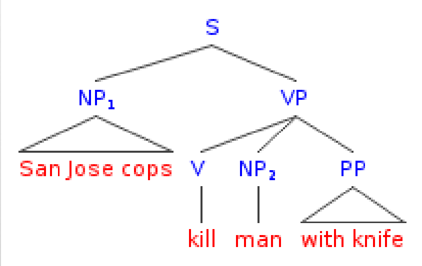
"Shots fired! Shots fired!" an officer said moments later.

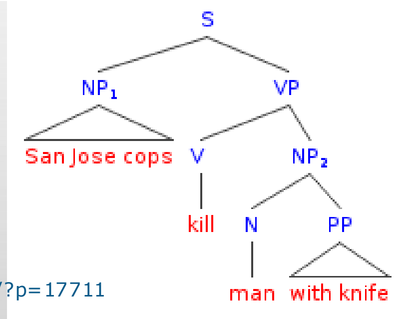
A short time later, an officer reported, "Male is down. Knife's still in hand."

Buchanan said she had been prompted to call the

Shoot continues on D8

Back Continue





Credit: Mark Liberman, http://languagelog.ldc.upenn.edu/nll/?p=17711

CHALLENGES -- AMBIGUITY

- AMBIGUOUS HEADLINES:
  - INCLUDE YOUR CHILDREN WHEN BAKING COOKIES
  - LOCAL HIGH SCHOOL DROPOUTS CUT IN HALF
  - HOSPITALS ARE SUED BY 7 FOOT DOCTORS
  - IRAQI HEAD SEEKS ARMS
  - SAFETY EXPERTS SAY SCHOOL BUS PASSENGERS SHOULD BE BELTED
  - TEACHER STRIKES IDLE KIDS

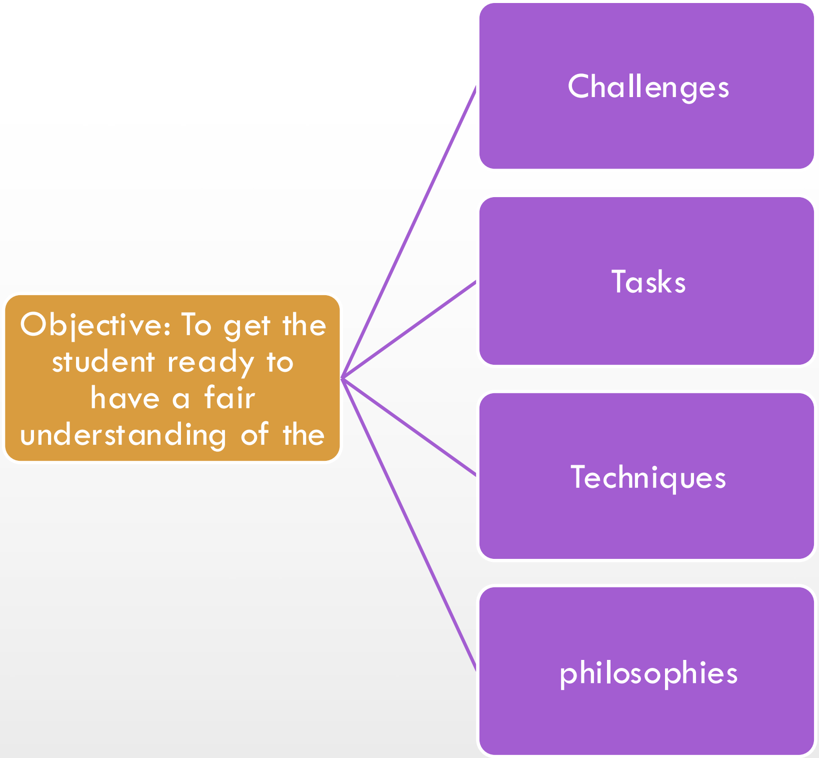
CHALLENGES – SCALE

- EXAMPLES:
  - BIBLE (KING JAMES VERSION): ~700K
  - PENN TREE BANK ~1M FROM WALL STREET JOURNAL
  - NEWSWIRE COLLECTION: 500M+
  - WIKIPEDIA: 2.9 BILLION WORD (ENGLISH)
  - WEB: SEVERAL BILLIONS OF WORDS

THE STEPS IN NLP

Morphology → Syntax → Semantics → Pragmatics → Discourse

WHAT WE DO IN THIS COURSE



FOCUS

- Textual NLP
- Address each level and tasks therein
- Identify areas of interest and build the groundwork